

Active Inference Insights 024 ~ Darius Parvizi-Wayne (Season Finale // Reflection)

Insights | Darius Parvizi-Wayne (Season Finale // Reflection) | 54:47

all right it is july 8th 2024 and it is active inference insights number 24 a reverse insight so welcome darius how are you doing very well i apologize everyone also for my stupid headband i'm currently in el salvador um it's very warm and my hair is going everywhere so i've just decided to strap it in so um it's a bit of it's a bit of a look i'm very well how are you daniel good well as we just briefly discussed this is sort of the season finale slash pivot point or next jump and we will discuss some of the inception and journey of insight series though one through 23 main episodes what is next for insights what is next for you and just however long that is so let us begin with yep insights how did it come to be and what were the beginning steps like well firstly it's annoying that we never made this to 25 you know it would be nice to have a nice square number but 24 that's life i guess um how's it come about well firstly thanks to your generosity um in many ways and the institute's generosity so i'm very grateful for that um i'd done podcasting when i was at university a little bit after but i think i just noticed there was a there was a sort of gap at least in the institute's offerings uh the live streams not like this one but the other ones that can be quite technical and i don't think necessarily offer a way in and a way to see how active inference exists exists in the sort of context of theories of biology or neuroscience or philosophy or whatnot so there was a space airport and i just i guess i guess it was also this idea that it was a way for people to learn alongside me um and i think that's always quite an efficient and effective way of podcasting uh because i don't you know there's one thing that where people know nothing and it's very difficult to have a conversation with an expert and there's another way that's two experts speaking and it's inaccessible for the audience so it was um i think we hopefully found a kind of sweet spot where i was sufficiently knowledgeable albeit to a quite limited degree at the beginning i'd say uh but enough that i could ask insightful questions and be curious the way it came about really was i did my masters at ucl in 2020 from 2022 to 2023 and about halfway through i sort of discovered active inference probably through predictive processing because obviously carl is at ucl and it's a big hub for active inference although i think there's an interesting relationship between active inference and ucl because not everyone's necessarily on board and then i started attending i got on the discord so miles who we just mentioned before miles tufty was my supervisor at ucl was also and is also interested in active inference and he recommended the discord and the institute so i started doing that started attending the textbook groups and then i think i just messaged you uh when would that have been sort of september october and it all came together quite nicely and rather quickly and then again just people to thank carl really first and foremost to be the first guest um a good three hours with carl was like a very nerve-wracking experience but i think a pretty you know throw yourself in the deep end and then i think hopefully it's just gone from strength to strength um and i'm sure we can sort of talk about how it's evolved one thing about that certainly by view count which is not everything the most popular videos on our

channel over the last year have been basically Chris Fields is 2023 physics is information processing and your series and now I'm kind of reflecting on that and uh when we have a presentation like a guest stream that can get technical or a live stream with a paper focused third people there it's human however it's also like a conversation with the literature and literature in conversation with other topics whereas in Chris Fields's case with the course like format that's a uh similar to a professor to audience and then what you kind of brought into play was like also the person to person and so that sort of like direct human element of the conversation which is much more like a textbook group but of course a textbook group still has the textbook group as reference point but once you kind of um swim away from the wall of any text then there's this like wide space of what you can ask people and and like how you prepared for sure and I think people naturally are attracted to the kind of uh you know what John Bavaker calls *dia logos* um this kind of mutual dialectical unfolding where I don't know we're both we're both um affording each other the opportunities for insights and one thing I always sort of wanted to do and never got around to do was doing in person insights episodes um there was the possibility of doing that I mean I think the more it can be embodied the more it can be enacted the more it can be situated the better and maybe that's the kind of idea going forward I mean I recognize also that one of the benefits of the institute is it's kind of its location is everywhere and nowhere it's this distributed map and that's really amazing because I think what you see on the institute is all these people coming from completely different parts of the world with completely different interests and um you being unified over this one of an interest in this framework at the same time I think yeah hopefully what what what sort of developed was this kind of mutual unfolding and I think that can really only be done be done in dialogue I think you have to be quite an exceptional speaker an orator to be able to do that just through through monologue yeah wow interesting point there okay so Carl and then dot dot dot dot dot dot

so yeah do any come to mind like what was sort of a um a lead-up or a follow-up that was surprising or you know without naming the innocence how how did people differ and how they approached like what you're asking did

well look everyone to be honest everyone was really amazing I mean for me just like being a pre-doc but being like to have the that generosity of people to give me their time and actually read and take it seriously and not give uh you know flippant or or oversimplified answers was really something and I think I recognized that with Carl when I asked him back to inference they gave me a sort of 15 minute answer which basically maths and I thought okay well this is not going to be this is not going to be a soft launch um so everyone sort of unified has taken it really seriously and been really engaged and that was amazing I think there were a couple pinch me moments um so one like I've always loved John for really since discovering awakening from the meaning crisis probably when I was about 17 or 18. he's always been a massive figure in my life and I was lucky to meet him last year and to be able to speak to him was a really bizarre and special moment so yeah John was really amazing I think also there were some people who I think what was nice about the process was integrating some people who weren't necessarily active inference per se but still had insights from different disciplines and different domains which

could bear on that to influence so i really enjoyed speaking to marilyn stenderer um about heidegger because this kind of notion of us being beings directed ultimately towards a futurity i think is very in tune with active inference and this emphasis on expected free energy but everyone's been yeah everyone's been a total delight i think also what's been really fun is having like the debate was really fun um about the mark of blanket trick and i think that's another thing to say i really enjoyed having people who weren't necessarily not only people from different domains but also people who weren't necessarily in the camp of active inference so whether that was you know a crystal lager for example just being able to probe the relevant questions i think is really important because the the the framework is always evolving and it's always important in my opinion at least to have these not necessarily outside voices but concerned voices just to temper uh proceedings and i guess you had that when you know you've done live streams of people like mel andrews um so i think you know if this something like this continues it's important that we break out and listen to other voices who might have something to say awesome thank you yeah well about the sort of contents what were some topical themes whether ones that developed like even over those months or just some emergent themes or like what topics did you find coming up or developing yeah so i think so i'm not by training a philosopher of science and so i guess when i discovered that to inference this whole notion of the map and the territory was not forefront in my mind um and in many ways my enthusiasm factor of inference i think has been quite efficiently and importantly mitigated or mitigated but just constrained by that conversation so to what degree can the maths actually really tell you tell us anything about lived well one lived experience two human behavior and three like what the brain is actually doing or the brain or the body what they're actually doing so that was a recurrent conversation that kept coming up over and over again uh this map territory or math territory um and i think hopefully it's been a contributor contributor to this wider conversation that i know is happening within active inference and the wider community again about what if inference can really say i attended a talk by tobias schlicht last week about well i think it was predictive processing's flirt with transcendental idealism but the conversation turned into a kind of map territory thing and i think a lot you know now and even people coming into active inference as a new field are very much aware of the importance of that conversation so that was one thing that i think was really really helpful the other thing that always kept coming up was consciousness and the lived experience and i think that's in part because that's really what my research focuses on and so people may have noticed that the podcast was slowly transforming into a podcast on consciousness but so be it um and in that case i think all that was cemented was the fact that i don't think we know very much at all um so i mean that in terms of the metaphysical question really the hard problem and then i guess on the flip side of that is the importance of taking lived experience very seriously and in my opinion always kind of starting with the phenomena doing the intricate and careful phenomenological analysis of the actual phenomenon in question before applying your models or running your experiments and so on so yeah the map territory consciousness um and and and much more and much much more and i think also just active inference is placed in the history of ideas um whether you are

coming from a kind of computational stance you're thinking about helmholtz machines or jeffrey hinton or its relationships in neural nets or whatever or whether you're a phenomenologist or whether you're an analytic philosopher of mine like thomas metzinger just recognizing that the history and the future is has come and will come after us and it's i guess always important to think about our place within that that stream of time nice but there was much more there was much more oh yes i'll definitely look forward to not just re-listening but also to jointly analyzing the core entities and the active inference ontology okay two questions one and two take as long as you want one what is active inference two why is it important to have insights about active inference so i think my wife i might have just put that but what's the first question what is active inference and the second one why does it matter to have insights yeah so an active inference is as we all say the corollary of the free energy principle it's um it's not the same as predictive processing and i think that's an important um proviso although a lot of people conflate the two so the free energy principle uh tells you in terms of mathematics uh that a organism or an entity that persists over time can always be described as if it is minimizing this upper bound on surprise what we call variational free energy um and then active inference i've always i i think to be fair there's an interesting meta question here about exactly what we are talking about when we talk about active inference but i've always thought of that as the process theory associated with free energy minimization now this as if or can be described as the rel is the kind of paramount thing here um because yes indeed there's a mathematical way that you could start from like a fock-er-plank equation or or or any sort of um start from your theory of physics which is that it's a um it's a whatever it is you know you have you have your mathematical equation and then you you say okay well therefore given this we can cast self-organization as this minimization of variational free energy the question is okay well we make the axiomatic assumption that things do persist over time so that's that's relevant um but then the question really is okay given that things are persisting over time and given that they can like that is a really nice launch pad how are things actually doing that and then i guess this is the overlap with things like autopoiesis and inactivism and cybernetics and all of this stuff why is it relevant to study active inference well i think because we we have to take seriously the notion that things are self-organizing um it's not a trivial fact right this is i think schrodinger's big insight um back in the 50s which was that life is a quite miraculous phenomenon it is this negentropic phenomenon that seems to defy the second law of thermodynamics um or at least offer a quite interesting counter example to this and there i know again this is not necessarily my ex field of expertise but i know there's a broad domain of thinking why that happens in the first place i mean why are these pockets of negative entropy in the universe anyway um and this obviously can take you all the way to stuff like arguments from design um all the way back down to like the role of lived experience so i think it's very important to study self-organization and but that said it's important to realize that the mathematics won't take you all the way there so for me i've always viewed active inference in the drainage principle as a very interesting launch pad but always with the recognition that the answer i don't think is is it can't be in the mathematics is the mathematics at least the way that i've always viewed it or come to view it as an instrumental tool

uh a very insightful instrumental tool but not as we say the territory itself and i know maxwell who i've just
chatted with a couple of days ago he's just released his preprint with dalton and carl and they make
very explicitly that point so i think the next steps in active inference is going to be something like
so what is so what is the point right like you have the maths and this is really interesting
descriptive tool but then what and is it is it really how secluded is it from what the brain is doing
because i think we forget that within the whole active inference ecosystem although it's different
you do get these ideas coming in like predictive processing or predictive coding or discrete state
space pom dp work and that's much more of a process theory about it's actually what is the brain doing
and then you know within those theories you start having the integration of stuff like active
inference uh sorry variational free energy and expected free energy so for me active inference
is really interesting because one thing seems to be true which is that we are self-organizing and
self-maintaining and self-distinguishing i think the real work needs to be done on saying
what is the instrumental tool what is actually happening in a realist manner and where are the where
are the where how do we get from one to the other and obviously this has been the paramount
question throughout but i think it's high time that someone comes in and really are
the the sort of the path it's like a generative passage but for the for the science
real active inference has never been tried well well yeah well it was a stochastic differential
equation which was which was escaping my uh attention um what that very interesting account um
impresses on me is you started with the downstream relationship of active inference to free energy
principles so that's the high road and in that sense there's a grounding in general thingness
particular partition more like a systems theory then you said that's a launch pad to get down to the
meeting point because you get that surprise or minimization vfe from the top and then with all
of these existing bayesian architectures as various low road paths those also have kind of convergently
described the evidence lower bounds multiple ways to look it up but it's like vfe is coming from it's
actually coming from the low road methodologically but it's as if it's coming from the high road so that's
one place they meet then you said well that is like that's like another orthogonal launch pad
for people who want to do something other than develop that infrastructure which will remain its own
research program and direction and everything but then it's going to be that so what so what okay so we
have rx and fur with the drag and drop and everything works fine and here and then what so that's that's
a very interesting and anticipatory i think i think just think first and foremost always for anyone coming
in just recognizing that these things don't all mean the same thing the free energy principle does not
mean predictive processing but it's a processing does not mean a you know it doesn't necessarily rely on an
upper bound on surprise and so on and so forth and so i think i think all credit to pa petzula and
friston for sketching this out quite effectively in their textbook and also the way that you guys teach
this in the textbook group and then also the the resources of the institute i don't you know it's
important i think i mean there are lots of twitter debates that rage about how ignorant the active
inference community is interestingly within the kind of ecosystem they're not uh and they're very i i personally
think they're very humble and aware of the of the limitations you know so yeah so so i don't know

what the ultimate solution and the synthesis is whether there is such a thing um but i think it's it's a kind of we've got to start the multiplicity before looking at the unity and then trying to see if there's unity unity in the multiplicity which is sort of both mcgill christen hey gallian all together you know and alluded to in the neat and scruffy also early in the textbook um okay so that that's very interesting um so several paths ahead yes first how about some of your own directions and then we'll talk more about like where could insights go sure so i'm starting my phd in october in at macquarie in sydney so i'm my supervisor in nez y politico and rafael millier and i'm very very excited and very grateful to have the opportunity uh and moving to the other side of the world as well which is i will be i am on the other side of the world right now but you know actually setting up residence it's slightly scaring but it's exciting nevertheless um and i want to explore principally subjectivity and the sense of self as well as what if like what a self is um so this some of this is foreshadowed in the work that we did on flow states so this was uh me lars carl brady yakub and um miles tuft and i think there what became really paramount to me was that the notion of self almost like the notion of active inference becomes conflated and confused because people don't make it clear whether talking about the metaphysical self as if there is a sort of cartesian ego or a sense of self and even when they're talking about sense of self they're not distinguishing between a self as subject and the self or a sense of self as subject in the sense of self as object so i i'm really interested in subjectivity the fact that experience seems to have a certain forminess and there's an ongoing debate about the degree to which that's a universal condition of consciousness so this is people like anna chau nica but really you're going all the way back to dancer harvey and um joseph parnass and louis sass all the way you know back into the 90s and i think linking all that together as well as an interest in being in the world in the heideggerian sense and skillful coping in the sort of drape for cian merle ponty sense so seeing if there's interesting connections between subjectivity and this familiar coping that we have in the world so that's a couple of strands and then i think naturally from that one becomes embroiled in what is being in the world to what degree does it involve representations there was just this big paper that came out led by johannes jaeger on the fact that relevance realization and agency and cognition therefore can't be computational so for me i read that and i think it's very merle ponty and very drape for cian um and so those are the big philosophers that kind of inform my thinking on this front so just seeing i mean it's all quite open but i think hopefully there's a kind of cluster there that's emerging because there is also a more meta question about the degree which we computationally model something like subjectivity if the fundamental aspects of cognition and agency aren't computational algorithmic and representational and i think it's a fair presumption that subjectivity might not be so this is a big mystery basically that no one knows where any of where the four meaners comes from and first and foremost no one really necessarily knows what the four meaners is so some people take almost without critical analysis that the four meaners is because there is a self a homunculus located behind the eyes for whom experience is given i think it's quite clear that that isn't the case from the human analysis from the neuroscientific analysis so there's really nice work by people

like kenneth williford um and going all the way back to angelo and sartré about the fact that consciousness might be the the subject with art of consciousness might be consciousness itself so it's exploring this kind of thing and then i then i guess for me as well something that's really important is collaboration um i've been really really really really lucky to be able to have collaborated already with really amazing people

um not only the people on the flow paper but also people like libby severs on the people uh piece we did on tourettes and self-modeling so really just trying to get integrated into this community this philosophical this theoretical biological community as much as possible and then seeing again what active inference can do for all of that

that so i think not necessarily making active inference the front and center because i'm not sure i don't think i could

without necessarily because of all of the big meta scientific questions that reside in active inference the map territory thing

so i think using it as a tool and then if there are developments either done by myself or other people about a more biologically realist robust theory of actually what the brain is doing but absolutely fleshing that out and trying to do this what maxwell and others have called the generative passage tying phenomenology or

the lived experience to modeling to experimentation but always always being aware of just how fraught with danger or how precarious and fragile that passage can be so that's the idea at least it's quite broad it will narrow down cool wow well that's it's exciting to begin the phd um what what you said about active inference not being front and center for some reason it made me think of like famous or powerful photography or documentation and it's like the map and the methodology so that'd be like the open source methods element however outside of the gear enthusiasts connoisseur respecter development community it'd be like okay but what about the picture so so not that that's a reductionist view on the totality of like photography or anything like that but just like yeah yeah it it may become relevant for for a larger and larger fraction to focus on how that picture looks rather than that's absolutely right what the methods and the tools are i think just i just think maybe because of the way i'm composed and maybe because my background is ultimately in some in philosophy of mind and philosophy i i i do think for myself starting with the phenomena is the first place never for not only never forgetting the phenomena but starting with the phenomena so i've just been writing this thing about attention and it's funny how strange the notion of attention can be once you lose sight of how it actually phenomenologically unfolds

and and how it structures consciousness so for me and again this is not disrespecting anyone who wants to start with the modeling or wants to start with the map but i'm probably inclined to go the other way that makes me think about textbook group it's like quite literally sometimes circle square line circle square yeah o s a b c d e and then it's like well even if you were going to make a map of your street and intersection you'd be talking about like 20 to 100 circles and squares pulled up to the level of abstraction of something like a pomdp and then that's like getting into the multi-agent and all of these concepts yeah and then also made me think of like what is this obviously great question to um to do the phd in

and the immune system and the sensory and effector and all these different self-concepts and then seeing what's

happening now with like agentic ai ai agents defined in terms of like perception and action action which is like one thingness and then at the same time as maxwell also emphasized in the uh discord recently like the agentic focus which is one system of interest or genre of system of interest actually minimizes the apparent scope because especially grounded more in the high road of just how things are as if they are rather than any particularly built architecture then it's like yeah there's no reason why it should be kind of an agentic boundary it could be like an ecosystem of shared intelligence and then that's where all these self-concepts are in like a kind of multiplicity almost by definition sure i mean i guess i guess ultimately

we don't know um so so like this notion of individuated selves is is in many ways uh at least i've always taken it to be an underlying axiom of active inference right i mean if one was to say that well we're all just non-individuated things in the soup then you can't really apply the mark or blanket partition in the first place so it's a healthy way to start but even if you read i mean i was just reading something yesterday on it was like auto poetic autonomic and activism um so just just it was a really interesting piece see if i can find who wrote it's a very it was very good um but ultimately and they were trying to do this saying okay why is inactivism why why are there being these two strands of activism sensory motor and activism and then autonomic are based on basically auto poetic inactivism and trying to kind of merge the two and saying okay well what's at the heart of living might be something like sensory motor and activism but not discounting the fact that there is this operational closure within the system but again one has to recognize there that like there's a there's a kind of embedded circularity which is that we already take the system for granted so we're saying like what what what cons what allows the system to self-maintain well it's operational closure but thermodynamic openness and then but what does that do well it maintains the self-organization and allows for self-maintenance and so there is and this is obviously recognized by those authors there is this kind of inherent circularity and this notion of like a strange loop has clearly always been pointed to at the heart of of intelligence and cognition so i think also getting to the to the bottom of that circle which for many philosophers would be problematic but actually for these theorists is kind of at the very heart of their theories is a really interesting thing but yeah again not forgetting that literally recognizing or believing theoretically that there are individuated things can look like an axiomatic presupposition and an active inference

it almost certainly is one so yeah i mean i mean what is a self really is the big question i guess there's uh there's really good paper i always do this it's like i'm in podcast mode again but it's um varela i think it's 1991 it's called a meshwork of selfless selves really good really really good because again it's not saying that this is a self in the sort of strict cartesian way that we all might have a focus psychology theory but like some of these are intuitive or viable ways that we could define the self whether this is a sensory motor boundary a cellular boundary and so on and so forth so yeah it's it's a surprisingly elusive concept the self yeah so much to say on that i mean even if you apply

a utilitarian or pragmatist approach to your own agency over an attentional profile over self-concepts it would be like that's the strangeness or part of the strangeness and the second order cybernetics with the observation okay i'll read some comments from the live chat and we'll keep talking about this before we do eventually talk about like where this is yes okay upcycle club writes greetings wave hand emoji i have really enjoyed the insights active inference series with darius i wish you all the best in your future endeavors well thank you that's very kind i mean yeah and then thank yeah go go go oh yeah thanks upcycle all right terry moldin writes i've loved the talks the latest one i really thought the free energy principle could resolve the different approaches if you remove the idea of a preordained agency of the individual make sense um yeah i mean again depends on how you define agency but um if you just define agency as the ability to again in like an activist tradition to set your own norms and standards with respect to your surrounding milieu um then active inference is very consonant with those ideas

[illegible]

I was very lucky, actually, as an undergrad to study Chomsky in linguistics.

And the notion of idealization comes up a lot in Chomsky.

And so I think I was, at least in that sense, quite well.

I know I said earlier that I'm not a philosopher of science,

but there were quite a few discussions about idealization in science.

So, yeah, in those cases, I mean, God knows how recursive and hierarchical these layers are.

And I think in some cases, all you have to do is point at it and say,

you know, at this degree of complexity, one must remain silent.

There's an interesting thing there about emotion coming after the phenomenon.

I'm not quite sure I agree.

I mean, I have flirted with the idea that, for example, in flow states, there is no consciousness or at least no emotion.

There is an ongoing discussion.

I'm not an emotion theorist, but I believe it's within appraisal theory about the fact that just if we take emotion to be an appraisal of, let's say, an interoceptive signal, it doesn't actually have to mean that this can only be done offline.

It could be done in real time.

And I think, informed again by someone like Dreyfus or Merleau-Ponty, that's quite right.

The emotion in many ways, whether it's...

Now, there's an interesting question here is about actually what the causal role of that emotion is, because there are two different things here.

One is its causal role and one is its temporal status.

The interesting causal role is where that emotion actually does anything for the organism who's trying to return to this kind of equilibrium or what Dreyfus would call your sort of optimal grip with the gestalt.

So I don't know.

I don't think the flowing agent is non-conscious.

I don't think they're a robot.

So, no.

Ultimately, to answer your question, I don't think the free energy principle solves everything by dispensing with agency, because, one, I don't think it can answer the question of agency.

Two, I don't think there's a robust definition of agency within the free energy principle literature or elsewhere.

And thirdly, under certain definitions, agency is inbuilt within actively inferring agents.

Note that I'm using agents with speech marks, because when I talk about agent, I'm not talking about some Cartesian homunculus.

So, big questions.

Big questions.

Which need lots of years of deep delegation to resolve.

Cool.

Okay.

More to the operational closure.

Yes.

So, having organized many, many live streams and worked with many people, what was very different about your co-organizing was basically you did all the invitation, you recorded it,

and just shared the upload.

So, logistically, what did it take in terms of research, background, prep, confirmation, all that kind of stuff?

How would you evaluate that?

And, or, how would you...

We'll just ask that question first.

Just, how was it?

How was it in how you actually did it?

Well, I like doing this stuff.

So, I love emails.

And maybe this is a betrayal of my age and inexperience, but I love sending emails.

So, well, you very kindly had the spread.

We have a spreadsheet, which is going to be hidden from everyone else.

But we have a couple of code of spreadsheet where we have just a list.

I think there was a list of people who had been on live streams before and then maybe potential guests.

And then we did another one where it was people...

And then there was a way of sort of categorizing it.

People I'd invited people.

I was thinking I'd invited people who had been on.

So, for me, I mean, the one was recognizing this is an active inference podcast.

So, starting with the kind of canonical figures.

That said, there were some canonical figures who did appear on the show.

So, that might be...

We could talk about that and who else could be there.

Because there were certain people who I think were meant to come on and then there were logistical timing issues.

And so, we can get to that.

But, you know, I think for anyone listening who wants to do something like this, people are unbelievably receptive and generous with their time.

So, really, ultimately, it was just a question of asking.

Of course, there are ways of asking.

So, I was always...

You know, one always has to be very polite and considerate.

And if people say no, which some people did, then that's totally fine.

But, really, it was just a question of asking.

So, we use Zencaster, which is a podcasting tool that only works on Google Chrome.

And they've been really good.

I mean, what's nice about them is I think they're a really small company.

And so, whenever I had sort of little issues, I would just ask them and they would get back to me super quickly.

So, shout out to Zencaster.

And then, I guess, logistically, the only other thing was just how big those files got.

So, a warning.

I mean, people who do podcasts and live streams will know.

I mean, these were big, you know, 24, 25 gig things.

And so, good Wi-Fi and an external hard drive turned out to be useful.

But, no.

It's gone quite smoothly, I'd say.

It's gone quite smoothly.

Don't get me wrong.

It's a learning process.

So, Daniel, you'll know this.

There's a...

Like, one starts with no...

I mean, right now, I'm just on my laptop.

But people will probably be able to notice the difference, right?

I don't have a light in front of my camera.

I don't have a different camera.

I don't have a microphone.

And so, I think it's an optimization thing, like active inference.

But it's sort of going, okay, I'm starting from here.

And I'm accepting myself from starting here as the complete layman, the novice.

And I didn't really set myself a goal of where I wanted to end up.

But I just knew I could get better.

And I hope, and people seem to think, that the quality of it has gone up and up as time's gone on.

But I think there's still massive scope.

And it's super cool that you guys now have charity status.

And hopefully, that sort of raises the ceiling here.

And so, yeah, I mean, personally, I'm just really excited to see where this can go.

Recognizing also the podcast space is a little bit saturated.

And so, I think when we talk about pushing this podcast forward,

thinking about how we can be innovative in setting up new schema for guests and live streams throughout.

So, yeah, surprisingly, quite a sort of fluent and not too tricky process.

How would you see it continuing?

Like, what would a very cool volunteer or intern or research fellow do?

Or how could they kind of jump in and continue just thematically and from the same intention center?

Like, what do you see that as actually being like for someone to do?

I think ultimately, first and foremost, I would always just say, speak to the people that you're interested in.

This is a sort of trope, I think, in podcasting, which is, you know, there's a vacuum going on.

Someone's going off.

Well, hold on.

This is the issue of doing this in an El Salvadorian hostel.

There's a lot of noise everywhere.

But I think ultimately, okay, you know, the audience is going to be, it's going to be quite evident to the audience if you're doing it for them.

And I'm not saying discount the interest of the audience, but I loved everyone I spoke to.

So I think whoever the researcher is, the podcaster is, ensuring that they know who they want to speak to is really important.

I think preparation was really important.

Now, I don't mean writing down all the questions.

You don't do this.

I don't mean writing down all the questions you're going to ask.

I always really had about three or four themes.

And maybe in certain cases, an actual, like, well-formed question, because it might have been slightly recursive and complicated.

But generally, and I always said this to guests before we went live, it's meant to be free-flowing.

So I think keeping that, at least for the podcast space, is really important.

I think that said, could the community actually be even more personal?

You know, one of the really nice things that you guys did was the interview with Carl, which I love.

I love.

I think it's great.

And so I think even little stuff like that could be really nice additions to the process.

Because, well, for many people, the only thing they see are their work and their Twitter accounts.

So kind of getting to know people, I think, is relevant.

And then, yeah, I mean, final thing I'll say is, I don't think we should silo ourselves in to only speak to people that we agree with, intellectually speaking, in this case.

And so constantly, you know, whether that's people who have actually been actively critical of active inference in writing in the past.

So, I don't know, Ezekiel DePaolo or Evan Thompson or someone like this, or even someone who you know might not just be a, even someone within the community, right?

Like myself, who is not dispensing with the map territory distinction.

Just having that dialect, that dialectic is really, really important.

So, yeah, I think there's plenty of that.

I mean, I'd love to see it continue.

I would watch every episode like an avid supporter.

And, yeah, again, I'm just super, yeah, I'd be very happy to help out wherever I can.

But for me personally, and we spoke about this, it's kind of time to lock in, I think.

I think this is one of those things.

I mean, I don't know.

I'll ask you.

I think at some point, to be totally candid, I was slightly worried that I was becoming a podcaster.

And I guess for me, what I want from my academic life, at least, is to be writing and reading and doing academic work.

So, I'm curious about your experience.

How much do you feel like the stuff that you do takes away time for writing, reading, and doing the kind of canonical academic activities?

That's a great question.

So, there's a lot to say on it.

And, of course, what I can say about my own last, like, 10 years doesn't apply necessarily broadly.

But during graduate school from 2014 to 2019, I was doing multiple weeks plus of fieldwork in the desert.

And while I was on Stanford campus, I was not taking video meetings.

I was not, like, in the digital.

So, during graduate school, absolutely, it makes sense to be really focused.

And then one thing I started exploring at the end of graduate school was live streaming some overhead view hand drawing.

So, with no audio.

So, that got me familiar with OBS, the open source broadcasting software, and just the logistics of doing something that is largely improvisational.

So, not preparing more than, oh, maybe here's a reference image.

Okay.

So, getting the time of live streaming as close to real time, which is this kind of, like, Marshall McLuhan angle on the whole thing.

Plus, like, people's feedback.

So, that's, like, one sort of lane.

Which, even now, doesn't take much more time than, like you mentioned, several emails and the time itself.

And then, where it starts to really be like, okay, this is, like, reading the paper, making the slides.

That, the question is, how slash is that aligned with, like, relevant learning and research formats?

So, like, the .0 style video that kind of came into play with just the pure background and context before, even if we had a broader discussion on the paper, that was, like, almost exactly the lab meeting slides.

Which, then, during the second half of the lab meeting, we'd, okay, go back to figure three, or, like, where was the abstract?

It's, like, being able to have sections up there on the slide form and have one person who could say, like, that they had really fully gone through the whole thing.

So, again, that's the conversation with the literature, but from a combination of alignments of education, research, service, interest, and then the logistical element of most streams not taking more.

But then, there is still that element of just, like, what has happened?

But, you know, that's its own thing.

But, Daniel, you're a beast.

I mean, like, it's worth saying, you know, I think everyone's very grateful for the work you've done.

I mean, it's, like, there's nothing else like it, really.

So, I mean, not to be, not to speak out of terms, but, like, well done.

Like, it's really, it really is something.

Thank you, Dears.

And I think also, like, to say that this is the place to go to learn about active inference.

I mean, I came to this, you know, I didn't, this, the institute is known.

And, I mean, the discord is getting bigger and bigger.

And so, yeah, it's really remarkable, you know, for me to be in London with not even necessarily over active inference people and them going, oh, well, I know there's a textbook group.

I know there's this chat.

I know that, you know, Daniel is doing this.

And that is a real sign of success.

And so long may it continue, I guess.

But, yeah, I think something you said there was, I forgot to mention, was all the paper reading.

So, I hope this is quite obvious.

Obviously, one has to know who they're talking to.

And so I think this in many ways informs my thing about speak to people you want to speak to.

Now, I'm quite lucky because I basically love reading academic papers.

But me, it's just a hobby, right?

Like, so it doesn't feel like work.

I love reading books.

I love reading academic papers.

I was also really lucky.

And I think this is just a logistical point that I had institutional access throughout.

And this is a broader educational point, which is a lot of people don't have access to papers if they're not open access.

So, I think that is probably the majority of the time goes into reading papers, reading the papers of the people you're going to be speaking to and making sure that the questions that you ask are really tied to their work.

Because there's no point in the book.

I think it's still crisp about quantum mechanics, really.

And so, I mean, I'm sure he'd have something actually quite miraculous to say.

But, you know, you get my point.

So, yeah.

So, yeah.

Maybe this is something that the Institute could also work on.

Finding a way for people to access papers.

Because for the vast majority of people who don't have a .edu, they can only see the abstract.

But, yeah.

But, yeah.

On the whole, yeah.

It's been a really, really splendid experience.

I mean, I think there are plenty of people as well that we could get on who would have lots to say.

Awesome.

Yeah.

I think maybe we could do something like see who's interested.

You could ask a few people.

And the time will be right when the time is right.

So, there's no need to jump on it, I think, to ferment and digest with, like, the 50 hours or whatever it is.

There's a lot there.

There's a lot there that people, I hope, will be able to glean.

Maybe not even insights from.

But just, you know, I think the role of philosophy in many ways is finding the right questions to try and answer.

So, it's not necessarily always about answering the questions.

It's about structuring the conversation.

And so, hopefully, we've opened up that landscape where the questions now are more concretized in people's heads, the relevant questions.

So, yeah.

Yeah.

I mean, I don't know.

Hopefully, an intern's watching.

I love this internship thing that you guys are doing as well.

And they're interested in taking it on because it's all theirs.

But, again, as I say, whoever's watching it might be interested.

I am, albeit, although on the other side of the world right now, not the most Wi-Fi robust place, although I have found some good Wi-Fi.

I'm always very happy to help.

Always very, very happy to help.

Cool.

We will see and we'll explore it.

So, thank you, Darius, for the very unique series.

Thanks, Daniel.

And I also...

Again, thank you for all your help.

And Blu and Ali and everyone who's part of the Institute.

There is, like, loads and loads and loads of work that goes in that people don't see.

And, obviously, like, they see my face.

But, yeah, you guys have been incredibly helpful and generous with your time.

And so, I'm really genuinely super, super grateful.

Thank you.

I think there's a piece there about, like, the future of content slash attention and whether and where we would really want a person.

Like, where I would want to know what you thought and where we'll be able to use synthetic methods to just, like, generate different kinds of material that would flood and potentially be indistinguishable or even preferable in certain ways than the real conversations, real dialogos that people would have in experience.

So, I think that, like, you know, the way that you kind of also helped frame it in that context is awesome.

So, thank you again.

No, I don't.

Although, just a final thing I'll say on that.

I think there's something very special about two humans speaking to one another.

And I think all I'll say is, without the appropriate embodiment, the appropriate social context, the appropriate enculturation and so on and so forth,

to have genuinely the effect, to have genuinely, not just mimic, but replicate the effect that can happen when the true dialogos is happening within two, you know, among two artificial bodies in this case.

I will, I will, I will, I will just say that I'm skeptical.

But in the meantime, let the humans talk.

And I'm sorry for sweating.

And I'm sorry for sweating.

It's 35 degrees here.

So, invariably, there will, there is, this is, this is not glamorous.

There is sweat.

Ah, and more.

So, thank you, Darius.

See you.

Thanks, Daniel.

Peace.

Bye.

Bye.